

seL4 - using the proof

Collection of possible goals for being able to "using the seL4 proof"

- **#1 Goal:** Be able to tell if a change to some seL4 file is breaking the functional correctness proof.
- **Source Code**
 - Which parts of the seL4 source code are considered for the proof?
 - And equally important: which parts are not considered (e.g. linker scripts, platform dependent stuff, ...)?
 - What transformation is applied to the source code before it is forwarded as input to the C parser?
- **Isabelle Code**
 - Where are the different parts located (repository)?
 - Can the Isabelle code be grouped into libraries and proofs on a module (file) basis?
 - What libraries do exist? Word arithmetic, refinement calculus(?), ...
 - What theorems, lemmas do exist?
 - How do these map to the higher level seL4 proofs (function correctness, security)?
 - Is it possible to create a nice hierarchical overview / grouping of all existing theorems / lemmas?
 - Are there coding guidelines being followed aiming for maintainable code?
 - How do platform details (RISC-V ARM-32, ARM-64) affect the functional correctness proof?
- **Papers:**
 - Identify the papers collectively forming "the seL4 proof".
 - Identify what aspect of the proof is covered in which paper.
 - Understand them at least to the detail to know what has been proved.